

ActInf GuestStream 043.1 ~ Tom Froese: "Irruption Theory: Enactive Account of Motivated Activity"

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hello and welcome this is active inference guest stream number 43.1 it's May 11 2023 and we're here with Tom Frost today we're going to be talking about the paper eruption Theory a novel conceptualization of the inactive account of motivated activity we're going to walk and talk through the paper and there will be a lot of time for questions so please feel free to add any in the live chat thank you Tom for joining and to you for an introduction and overview okay thank you um so yeah hello everybody um my name is Tom Frost I'm the head of the body cognitive science unit at the Okinawa Institute for Science and Technology Graduate University in Japan and um the scientific mission of her unit overall is to take the advances the theoretical advances in cognitive science that are sometimes brought together under four recognition so that's embodied embedded extended and active approaches to cognition which to our minds are on the right track in terms of how we should think about the mind and how it relates to the world but so far I haven't gotten a lot of exposure and experimental work um and so our unit is basically trying to become an interface where we can do exciting work on the theory side of things and then try them out and experimental work and we have work on human computer interaction we have um uh EEG brain Imaging we do sensor substitution research um there's all kinds of stuff about behavioral coordination and social settings and haptics and um and so um it's very exciting place to be because we're kind of sticking our necks out a little bit and saying you know do all these cool ideas that you know love and cherish actually make a difference in how you know science is done in the lab and that's you know that's why we're here but at the same time uh what what that kind of ambition makes clear is that some of the theoretical work also needs to adapt and change and and Advance such a more directly connects with the kind of things that we can do experimentally and so the current paper that we're going to discuss is one of those outcomes where I'm trying to think a little bit more clearly about how we can operationalize some of the ideas that are coming out of the inactive approach such that we can start thinking about how to formalize them and how

to even measure them using something like you know analysis of EEG signals awesome very cool theory and practice and um there's a lot of prose papers and a lot of philosophy and so it's just very exciting even with the Technologies you mentioned like how this really comes into play and how it does make a difference so what's the story with this paper how did it come to be and where does it fit into that broader work of the group yeah sure so um let's zoom out just a little bit before I go to the specific paper so for most of my career um I've been working on work like because like Origins you know origins of Life origins of mine origins of social cognition origins of social complexity um and I think that kind of look at Origins is share quite widely among my colleagues so what we want to understand is what are the conditions the necessary and sufficient conditions for there to be a living being or there to be Agency for there to be sense making or participatory sentence making um and so for most of my career I've been looking at other you know developing ideas about what are those conditions and in the background there was always this assumption that that's pretty much all the work that we need to do in order to have a cohere notion of let's say agency um in the sense that uh Notions such as active versus passive um or going back to avano's famous book action and perception which was like you know the super hot book when I was a PhD student and it was discussed so widely we never really questioned what is the action part of this uh so how do we Define something as being active versus passive um yeah there are some ideas about regulation and normativity things can go wrong and right when there's action in play and intentions in play um but to really think about how those intentions and values could make a downstream difference to let's say the neurophysiology composition of the body is something that we just left to the side and didn't even really consider us part of the problems that we had to solve so um somehow uh doing the attempts at modeling this uh so another stream of my research is an agent-based simulation model where we try to look at the minimal dynamical conditions that lead to a certain kind of behavior so let's say I can ask what is the minimal setup that I need in order to study social contingency where One agent is sensitive to the dependence of the others movement on its own movement and I can do that and you can analyze that and I can evolve it and I can study the state space of it um and and then we we thought kind of like our job is done you know we've understood the Dynamics of it and now let's move on to the next problem um but something started bugging me about that when I tried to more integrate that kind of work with my interest in phenomenology and this goes back to uh Francisco varela's work and neurophonology where he proposed that dynamical systems theory could be the kind of formal bridge between on the one hand a kind of Australian style analysis of immunology

and on the other hand empirical work in Neuroscience primarily but also including the body and interaction with the world and so his idea was that because dynamical systems theory doesn't make any ontological commitments about what it is that's changing in time all it is doing is describing how something is changing in time it can equally be applied for mental processes which change in time and physical processes which also change in time and then the the challenge is just to find a kind of general form or structure which integrates both of these processes changing in time and voila we've solved the mind-body problem you know I'm simplifying but that does kind of like the general ambition right so um I'm not a neuroscientist by training so so I did the best thing I could which is work more at the level of modeling the behavior of systems but again the ambition was if I can create a dynamical systems model of something like social contingency then on the one hand I can connect it with something happening in feminism and on the other hand I can make a hypothesis about what would be necessary or sufficient in terms of experimental work the problem is that that kind of work after a while made it very clear that um the dynamical systems approach didn't leave any room for anything else to make a difference other than the equations that I was putting into the model which is to say that you know as long as I knew the initial conditions of my simulation and as long as I knew the equations that determine how stage change over time all the rest is already predetermined I can exactly replicate again and again by just resetting this relation so it unfolds over time in a predetermined way sometimes we cannot predict it in advance so it's not necessarily predictable without you running it but it is repeatable and identical way so there's nothing happening in each time steps so to speak that could ever be surprising once you've seen the whole thing unfold and so that really started bugging me because why I got interested in the inactive approach originally is because it was all about action and perception it was all about agency it was all about intentions and sense making that meaning makes a difference to how we behave in the world um uh you know what happened to the values or like you know we were always talking about that you know the inactive approach has an account of how values can be intrinsic to the organism because of its precarious self-production and this goes into the direction about the creases and so on but here I was confronted with the with the fact that if we describe all of that in dynamical systems theory there's actually no room for values to make a difference everything is already predetermined by the Dynamics um and so in unpacking that dissatisfaction a little bit eventually resulted in a complete turnaround where I now reject um the adequacy of a purely dynamical systems approach to describing phenomena that have to do with agency and sense making and meaning

and value which in this paper are basically a group under motivations there's something else making a difference to how the agent is behaving over time so so that's the kind of like background story and uh I have to say that it took me maybe three years to get myself to this point because it's a big break in my own thinking and what I've written in the past and um you know at this point I'm not sure how many of my colleagues are actually on board with this change um so it's kind of like exciting and nerve-wracking at the same time to say like you know mid-career changed that suddenly throws out a lot of what has gone in the past but uh you know this is why we do science right sometimes you have to change your mind and so what we can do today is that I will talk to you a little bit through this paper and you can probe it and uh you know who knows eventually I might change my mind back and say actually this is not correct right but right now I think this is you know the way that we should move forward and there's a lot of rich like implications to further unpack and so so yeah personally I'm super excited um that that you know that I had this change of mind about how to think about the relationship between mind and body um and I think it's going to be um you know helping to some extent inactive cognitive science or embodied cognition more generally to live up to its own expectations that it has something serious to say about how things like meaning and value make a difference to behavior wow is that right for the context should I jump into the the paper oh it's it's quite a saga thank you for sure sharing that really helps a lot so into the paper we go okay all right um right so maybe clarification so motivated activity by the way can you see my cursor or uh yes we do okay all right so motivated activity is the term I'm going to use um to capture all of these other aspects I was talking about so meaning value intentions it could also be higher level Concepts like beliefs desires so anything that has um a kind of like Asian level uh Association um and which is not just uh if you use Stuart Kaufman's terms just a mere happening so something is doing something uh and that kind of brings with its normative criteria of either succeeding or failing or being good or bad at whatever it is that needs to be done so it's a kind of quality of the difference to just mere activity um so because I was you know I need a term that captures all of that vast diversity of possible normative behavior and then here I'm just gonna say the activity is motivated by something else and that something else will be one of these uh agent level uh Concepts nice so it's supposed to be a very broad Theory it just makes me think of how an enzymatic reaction could be catalyzed but it wouldn't necessarily be motivated whereas motivation for an agent could be a catalyst yeah yeah so uh and here I don't go into this but the kind of in the background there's the idea that um the next one of the next steps will be to generalize this

into a notion of biological regulation um and so there's a lot of Life Sciences happening here at my University um and if you talk to them you know they they basically treat their systems as mere happenings right it's as it was just a pretty physical system without any motivated activity but I'm not sure that that's the right way of thinking about it so when a cell does something for a reason like up or down regulating things possibly this account also applies there because then Notions of success and failure and how much and how little is you know the right amount and so on start making a difference of life and death sometimes and so we should expect something similar to be happening even at that kind of low scale level so although I'm gonna unpack it now in terms of the human level because that's the most intuitive level um I think that the general principles in the background apply to all kinds of Regulation even unconscious biological resolution all right and just I'm sure we're gonna go into it but what is the etymology of eruption right I mean we can already uh get into this uh now we're given that we're already kind of uh into the kind of contextualizing part um so some of the influences that uh have been very impactful for me uh go back to cybernetics and particularly the work of Ross Ashby um and he has this notion of ultra stability which is supposed to be a way of explaining how can a system uh become stable after being perturbed out of that stability in a spontaneous self-organized way without actually having any kind of like top-down explicit knowledge of what it is that's going wrong or how it should do to fix it um and so again it's supposed to be a generalizable account of biological regulation um and you know most work back in the day was taking the opposite extreme of saying Hey I want to have perfect information about my target I want to have perfect information about my error and then I want perfect information about how I should reduce that error in order to hit my target right and that's kind of like control theory feedback based control theory um and we can engineer systems like that but nature is messy and a lot of the times you know whatever hits the system you don't know what it is that's hitting it you just know that something is not quite right and so Ashley basically proposed that as long as the system is able to break and this gets close to eruption um that break basically means a change in its structure or a change in its organization and that basically means that as long it's not dead after that break it has a chance to have hit on the configuration that actually is a the right kind of response to whatever is perturbing it it's a new kind of organization so it could be a stable one and if it's not stable then it will break again until it eventually hits another configuration and which it doesn't break anymore and it has basically found a new pocket of stability right so it's kind of like a example of stochastic search um but you know the the interesting part is that for Ashby um this break

comes from the outside it's like you know he has this example of a kitten being burned by by the fire and it's far it touches the fire it you know it gets burnt and it has to learn not to touch the fire anymore right so if there is no breaks happening according to Ashby the system wouldn't do anything it would just be static to some extent it would just maintain its identity and sit there and so something perturbs it in the right way and then it would react and so um for me that was always kind of like a very strange way of looking at life I mean that view maybe applies to something like a computer that if you don't touch it it just doesn't do anything and then if you do something to it it will respond but living systems are somehow intrinsically active right so they're always doing stuff uh they're always changing adapting developing evolving learning um right and so uh the idea that they are just uh that all of those changes are only responses to the environment rather than something that is intrinsically driven doesn't make much sense to me there must be also something on the inside of the organisms that makes them Restless in a way and so one of the early attempts at getting at that was to think of a notion of self-breaking what if actually systems continuously break themselves a little bit which basically then has the consequence that the organization slowly also changes over time and so then they are no longer dependent on their environment for adaptation but this basically becomes an intrinsic source of adaptation and so this notion of eruption goes back to Latin which is the kind of like sudden breaking into or meaning of breaking actually and then kind of internal break in the system all right so into the into the paper um so like I said uh is you know the main motivation is getting a grip on what do we mean by action um and like that's one of the concepts that goes back to the 90s right I mean like if you think about the embodied mind the classic text by uh Varela at all um you know the way they Define uh the inactive approach is basically by linking action and perception into a loop um but what is action uh can a robots perform an action uh is it enough that there's just mere movement um and this is a long debate also with ecological psychology is it necessary that this movement originates in the organism or could it also be something that happens from the side of the environment as long as it's a relational change and so on um and so basically uh I I argue that I don't have a really good grip on what do we mean by action um when and when an agent makes a difference I review the inactive approach um not sure how much I should go into this maybe just to say uh briefly um so this is what I meant in the beginning where most of the work of the inactive approach has focused on the emergence of living beings with an intrinsic perspective of concern um uh with sense making capacities and so on rather than the downstream consequences of when that is actually in place but just briefly the idea is that um

when we attribute purpose to an artificial system we can do that right the function of a car is a drive and if it doesn't drive it's broken so there's a normativity there we have a criteria for evaluating if it's doing the right or the wrong you know function um no organism seems to be of a similar kind we can also look at an organism and say you know is it behaving appropriately or not giving it circumstances in its environment and if it's not uh well that's pathology that's a disease that's sickness and so this is actually you know at the core of medicine to some extent if he didn't have any criteria for distinguishing when things go right and when they go wrong well how could we ever try to heal someone right so intuitively we have we're working with these Concepts already um so the inactive approach the idea is that basically the main difference between a car and an organism is that the car is produced from the outside whereas the organism produces itself from the inside so there's a network of processes of material production which basically create and synthesize all the molecules that compose our bodies and those molecules of our bodies are exactly the ones that permit the reactions to proceed in building the body so there's a kind of self-referential self-referentiality in the way in which we are organized and because of this self-reference the functions that we that the body is doing are actually created by the body itself that means they are intrinsic to the body and so the idea is that this is kind of the way in which we can move from just a purely external attribution of value and purpose and you know classic teleology to some extent to understand intrinsically based one because of the self production of the organism and there's one additional Point here so this would be a way of saying the goals are intrinsic but then there's also the question like why would anything matter to the organism in the first place right and this is another important difference between life and let's say machine like a car if the car doesn't work it doesn't care about that it doesn't work right it's just as happy if I can say that to not be working compared to working it makes no difference to the car very different for organisms they will be processes uh coming in active when things don't go right right so in order to try to compensate somehow and so the idea is that because the living organism's being it's very existence depends on these active processes that's why things matter to it it's because the organism itself is its own creation so to speak um and and not something that is just created from the outside um and so this is why precariousness as a notion is so important for the inactive approach that given that it's an active process that's taking place under precarious condition which means that if the processes are stopping or not proceeding in the right way the entire being of the organism would just cease uh that means the being depends on the actions of the organisms and hence the outcome of those actions matter to the organism that's why it

has an intrinsic perspective of concern on its environment and that's where meaning comes from that's why the World shows up as mattering to us and to organisms right so that was a kind of mini summary of I don't know two decades of inactive work on on um you know sense making so um did that make sense to you otherwise I can go a little bit further here but um I think this work has been you know much discussed already so maybe we yeah we can we can we can return but please continue so uh here's a couple of diagrams of what does this mean in terms of thinking about uh how value could make a difference so if you've got this uh paper from I think back in 2010 by the part of it all where uh once you start thinking about value or something intrinsic to the organism and as tied to its process of cell production you start to see that how other people use the notion of value in cognitive science can be quite different and so the diagram on the left where we have a value-guided learning shows the way in which many architectures in cognitive science think of value which is basically is a kind of evaluation system so there's a kind of sub-component somewhere in your cognitive architecture and the task of that subcomponent is to make some sort of judgment about the situation in the world and then pass the outcome of that judgment to in this case the behavior generating system and so the critique that the authors do of this is to say well you know you can call this a value system but you could call it anything you want to some extent you know you can call it like uh um I don't know decision system you can call it uh um a comparator system or or whatever um but there's nothing intrinsically different about the functions that it does to some other function in the in this architecture so there's nothing qualitatively different about the kind of behavior the sub component realizes and it's very localized so there's a worry that maybe what's going on here is that there is an unjustified verification of value into something concrete and specific inside this component and so their alternative proposal On The Rise is that value is something that emerges out of these self-producing self-regulating processes of the organism and somehow shape that those processes uh in the background so I think this has been kind of like the standard way of thinking about value for maybe the last 10 years or so on the inactive approach and it's something that you know I was also in that tradition and now I'm just one way of thinking about what I'm doing now is to really ask the question well even if a value would be an emergent property rather than a local concrete property there's still the question of how does that value actually make a difference uh to the processes you know if we remove this word value from our diagram would it make any difference and so what I said before about my simulation models is that I mean I don't see any difference right so I can just run my agent-based simulation models and then I can say oh this emerging property here I want to call

that you know the value of going towards the goal or something like that but the system doesn't care about that attribution I mean you know I could also not call it that or I could call it the value of being happy and the behavior of the system would be exactly the same it just makes no difference to the Dynamics of the system and so what what are the options uh you know how do we integrate this notion of value into our scientific account of behavior just just one one note on this um on on figure one um the left side is very compatible maybe even isomorphic with the reinforcement learning computational architectures where you have some behavioral selection module like you mentioned and then there's like a secondary evaluation that's like annotating or layering over the process of behavior being generated but at least value plays a specific semantic role whereas as as um shown here it values everything then it's nowhere you know if it's in the foreground it's in the background and um you have a closed loop and you have no need to explain anything further so you might as well just add all kinds of you know draft manuscript not for circulation and this is how organisms work you can add all kinds of things but that's not going to break the closure of the self-sustaining dynamical processes in the world and that closure has been highlighted by um a lot of the work you've described recently as well as Kant and others who are described in the manuscript yeah so um that's right and like you know when we ask ourselves why is the inactive approach of value not more successful even though it's somehow compelling in its biological grounding I would say it's precisely because it doesn't give a better account of the efficacy of value right so yeah maybe the the traditional account has a set of problems such as like you know where do these values come from in the first place you know if they can just be captured by functions you know do we lose somehow the qualitative aspect of them and so on but at least they do work right I mean at least they have a function in the architecture that's very specific and formalizable and can be implemented in these kind of for example reinforcement learning paradigms um whereas like the inactive notion is kind of diffused uh and almost verges on a kind of nihilism if you really want to look at that so it's like if the if the value is identical with the global dynamical properties of the organism well that identity relationship doesn't allow it to have any function in its own right you know you could just describe the whole system purely in terms of its Dynamics um and so it's kind of like a slippery Road what they want to do is shift the notion of value to the level of the agent so as a whole the agent uh you know has an intrinsic perspective on the world that means that value shouldn't be for some sub-component inside the Asian that's the kind of motivation of why why you don't want to go with the kind of classical View but then the new view where you bring in this agent level

perspective with an intrinsic concern for its own supervision and about the world um what's the scientific account of that well you've removed value now from the sub-personal story and put it in the personal level but you haven't given us an account of how the person on the sub person will actually connect with each other all right and so so basically you've actually just like voided the kind of sub-personal level of any kind of relevance in terms of connecting with something on the Asian level so yeah so so I think we're on the same page so like yeah one way of thinking about eruption theory is to say uh we need to do more here that you know if we are wanting to make a place for value at the personal level such that we can say you know I am talking with you because I want to right there's an agent level kind of characterization of what's going on then we still have to tell a story about how that Asian level wanting something actually connects with all the material basis of our Behavior such that it realizes that intention and that's something that hasn't been done uh you know there's like these are two different levels of description and yeah there's no representations and no values at the level of the empirical story at the material basis and everything happens at the level of filmology um but yeah there's a disconnect between these two stories so far yes and it's not the same thing as the hard problem or the question of awareness it's a causal relevance and the image that came to mind is like a buoy in the ocean and we explain its uh Movement by appealing to the flow of the waves and then uh making a model in which a sophisticated buoy also going with the flow is able to navigate and surf or something you've explained away Surfing by a really delicately balanced set of channels such that the flow is one of the same as surfing in that context but that gives no causal relevance to wanting to surf and so one is either left with this just a happening view of wanting exactly or to just say no it never mattered yeah so your your example is quite close actually to the kind of original materana example of structural coupling uh where he treats people or organisms more generally as a boat drifting on the sea that's being pushed by the waves and the winds and so the direction of the boat is not directly determined by the wind hitting um you know imagine a sailing board right so the Sails are up the wind's pushing it but instead of going in the direction of the wind what it does is go sideways right and so that's his example of saying structure determinism which means there's a certain kind of autonomy of the system with respect to what impinges on it from the environment um but you know that's one thing but the other thing is to say that there's absolutely nothing happening there at the level of intentions of wanting to go that way or that way or that there is a function of wanting to sail in the first place and things like that there's no room for that in that story um you know we are all like like you know uh just drifting so to speak and that's this concept of

neutral uh drift that they put forward so um yeah so we're we're basically yeah so this is like the you know the kind of conundrum that is faced by this kind of biological biology of cognition and active approach but I mean also of employee cognition more generally um so the question is how we can do something different there um let me see maybe just to unpack a little bit more of the the inactive approach with this diagram this is the kind of uh because we're bringing in biology of cognition just now and auto courses so maybe it's good to just say a few words about how the electrical push differs a little bit from those previous Traditions um and so the classical view of Auto Grace is by Mata Rana and varila is about cell production um so that means there's a system that creates itself and its own identity where identity is the defining organization of that system and that's coupled with this environment and that's all and that's it and so as long as you don't die you're just gonna persist coupling and self-producing uh and if you die well that's it um and so just because of that there will be a negative selection that if you're not particularly well coupled with your environment you're not adapted you will not persist and so that's the whole story of biology recognition there's no level of value or intentions or agency or experience these are all kind of narrative redescriptions but that don't have any kind of cause of relevance for understanding the real behavior of these systems and so enter the inactive approach and they say no we need to bring an intrinsic goals and concern perspectives um and this has to do with the finitude of the organism and the precariousness and and so on um and then for a long time it started feeling like uh we're kind of like starting to create like a shopping list of properties again like you know it has to be self-producing it has to be adaptive it has to be precarious right and that's exactly the kind of thing that uh Alto quizzes didn't want to do he wanted to do a kind of first principle operational definition of life rather than this classic shopping list of what it's like where you basically list your favorite properties of living systems and then say you know if a system more or less satisfies most of these properties I'm going to call it living right so it started going again into that direction and then there was this nice work by the Apollos and called it in recent years where they tied Auto places a little bit more closely with the thermodynamics of it and that meant that you think about what Auto places needs to do in a material context on the one hand of course it needs to maintain its kind of structural Integrity so it needs to be closed off somehow from the environment in order to be able to regulate its interaction with the environment but he can't be completely closed off because if that was the case it would just like starve itself to death you know it needs energy to do the work but that energy comes from the environment right so actually it does it needs to be open to energy Flows In order to be able to do the work that

needs to be Integrity maintaining so there's suddenly two conflicting needs here on the one hand it needs to close itself off as much as possible that would mean it would be as autonomous as possible and yet at the same time to maximize the energy it has to do its work it would actually have to be as open as possible to these flows which would undermine its Integrity so the life is faced with this weird primordial tension of having to satisfied two needs that can never be satisfied fully and never at the same time and so then the key move is to say that this is the reason why life is precarious intrinsically it can never be in a stable situation and the only thing it can do is regulate album open or how close it is over time and that this regulation then becomes the basis for agency that's why life is different from other kinds of far from equilibrium this this reminds me by analogy to attention-based learning where just um dilating and paying more attention or constricting and paying less attention is uh able to be part of an Adaptive strategy for dealing with streams of incoming noisy or partial information and maybe one reason why um attention feels so tied up with agency and so precarious as well because it's it's um walking that impossible tightrope and it knows it can't be on either Extreme as you pointed out they both result pretty rapidly in death so we're only going to come to see organisms that do have a dynamical navigation of this tension yeah yeah so actually this is a general principle I think that you know this is the the origin of it or maybe like the the most basic form of it but I think it's true of everything to some extent every time there is a way in which we're coupling with the environment there will always be a tension between doing things your way and actually at the same time having to be open to the environment um and like another kind of area that's interesting in this regard because we're doing EG hyper scanning in our lab is that most of the time when people think about hyperscanning they think just about the coupling so the question is how much interpersonal neural synchrony for example is there in the condition but what they don't think about is that actually you know that probably what's happening is that sometimes we are more strongly coupled and sometimes we are more you know distant disengage more coupled disengage couple disengage if we were always super coupled that would be kind of pathological I think you know it would be very hard to maintain that and it would feel very uncomfortable and you wouldn't get anything done so so actually a sensitivity to this kind of tension actually has methodological implications down the line because we still don't have to ask questions about the diacronic changes in the strength of coupling of the organism with this environment and if you ignore that and just take the average let's say we might actually completely miss the phenomena what other just analogy there is with so-called futile Cycles in Biochemistry where you'll have like the

forward and the reverse reaction happening at the same time and people sometimes characterize that as wasteful yet when you pull back it's actually the coexistence of these two Pathways that allows adaptivity and so it's like the paradigm shift is not to see it as wasteful but adaptive it's just simply adaptive it's simply valuable for the cell to participate in that um bi-directional activity right yeah and just another example sleep all right so in our culture sleep is not really valued right you know energy drinks are a huge industry you know it feels like if we could engineer sleep out of people then you know people would go for this and you know I would make a lot of money with that but the question is you know given that sleep is a really vulnerable state for the organism and we find it across all animals to some extent uh it must have really important reasons that it wasn't selected against right so even though probably the majority of animals die when they're sleeping let's say right because they they don't they aren't aware of dangerous and predators and so on still we're sleeping you know for a large chunk of each day why right uh and so it's this thing like you know you need sometimes you need to decouple and if you don't decouple you're just going to dissolve basically you know think about sleep deprivation you know if you do too much you actually physically die of sleep deprivation so it's not just you know you know down time so to speak it's absolutely essential for proper functioning and so just bringing it back to the paper one way of thinking about what's going on here is that it's not just that we are opening and closing uh you can ask like why if it's not a metabolic reason what could be the reason for this um and we've got a companion paper for this one that's gonna come out and buy a systems hopefully soon where we have a little model that shows that if you push a system far from equilibrium make it an open system basically and then close it again let it fall into its equilibrium State and then repeat that over time and at the same time you add a little bit of historicity in the background which is the kind of sedimentation of the states it has visited in the past such that it's more likely to return to states that has visited in the past what happens is that the system starts building up an associative memory of the States forms it has visited in the past and can actually then start generalizing two forms of stability that has never accounted in the past and that if you didn't have this kind of setup would never in principle encounter because it's just too hard to find in the state space and so we basically redoing here the kind of agency diagram that I just explained and we're just making explicit the historical dimension of this process is that if you're switching between far from equilibrium and equilibrium modes and there's a little bit of historicity here that's really when agency emerges because the system then becomes able to jump to new configurations that would have been otherwise practically

impossible to reach for it so for me that's another another story so you don't you don't need eruptions to to think about this but it kind of fits into the same um you know larger rethinking about the you know what's the role possibly of of breaks actually and that's all like that's that's the bigger picture I think we should you know allow much more room for irrationality uh indeterminism noise uh you know when things don't work the way they should and just say that that's a necessary compliment for things going well I think that's if that could be like one of the take-home messages of our conversation yeah and that reminds me actually of a lot of empirical work on heart rate variability uh gait gate variability and um one Viewpoint the car Viewpoint would be well you want those uh steps to be really even length you want the heartbeat to be like a metronome you want the breathing to be like a metronome but in fact it's the opposite that are associated with all cause um mortality such that higher variability patterns are associated with um more vital Health that's right yeah yeah um and uh I mean there are good reasons for that right so there's this whole control theory aspect to it that we want the controller to have higher degrees of freedom than that which is controlled and so you know at heart rate variability is important because you want to keep your blood pressure constant no matter what kind of situations you put yourself into so basically the heart has to take up all the the slack so to speak of the variable conditions of me getting up from the chair and so on and the blood pressure has to stay constant right so that's why you have a lot of variability there um but there's the more General notion of thinking about why is variability associated with health and then maybe eruption Theory comes in again a little bit in that as we will see because it's basically giving Arguments for saying that signs of agency making a difference might look like increased variability in the system so to get there let me just start to get a bit more narrow and what is the eruption Theory trying to achieve and so and the next section of this paper um uh what I'm trying to propose is that the challenge that is lying ahead of us is thinking of developing a motivation involving account of motivated activity so that the reason why I put it like this is that um I think the large majority of work on motivated activity and again remember I'm I'm using motivated activity as an umbrella term or Asian level Concepts such as intentions beliefs desires experiences you know goals um you know so and you pick your favorite let's say intentions right or Consciousness for that matter um that you want to say that Consciousness makes a difference for what you're doing are your intentions well what what are the standard ways of accounting for that um you know we can say that in terms of Consciousness you know maybe we can go with Walter Freeman's approach which was quite influential for the early work of Barilla and Thompson on neurophonology and

Freeman says well Consciousness um you know we've got this order parameter in the brain uh which kind of like organizes the you know the firing activity of all the neurons and so given that that's a high level property immersion property that has Downstream effects and we think that Consciousness is a kind of Asian level property that has Downstream effects so why don't we say that Consciousness works on the brain through the order parameter okay so so uh uh that's you know let's say that's one possibility right let's think about intentions we can think about intentions again in terms of dynamical sincere many people worked on it uh Juarez has a book on it uh Deacon worked on it um there's a whole literature and philosophy of of action that applies a kind of dynamical approach to saying well what what intentions do is uh they Place constraints on the degrees of freedom of system such that it realizes the action that is you know needs to take place um and so it's basically emerging constraints that have top-down effects on your degrees of freedom so kind of similar to the order parameter idea um uh but my main issue with these kinds of accounts and there are more of them is that they kind of offload or Outsource the efficacy of the higher level property to something else because I could also just tell the same story let's say of the order parameter without talking about Consciousness at all I can say look the you know I'm going to be a complete reductionist physicalist and I want to understand how the brain works I could still tell the story of hey there's all these neurons interacting there's an emerging process that has top-down constraints and how the neurons are firing nothing there commits me to saying anything about intentions or Consciousness so then what's the point like why do we need to appeal to Consciousness in the first place we don't get any explanatory Advantage you know the the neuroscientific story doesn't get any extra advantage and also if you wanted to explain something about how Consciousness work well you're also not saying anything because you know it makes no difference to the other story and so that's the kind of situation that we're in so that when I say I want a motivation involving account I want something where the motivation itself as such as a motivation makes a difference to the system and not just because I'm identifying it with something else like an order parameter or constraint right so I think if we don't have this ambition then we're basically missing the point I mean this is gonna this is really the kind of main challenge that I see facing us right and so eruption theory is one attempt at solving this problem and in more recent work I'm gonna kind of try to you know crystallize this out as the hard problem of efficacy how do we make sure that our Asian level properties that we care about actually our thickest in the world and not just because they're super being on or identical with or whatever you want to say with properties that are not actually at the Asian

level that's that's to my mind you know that's that's where this work is situated and if you can't solve that I think it's a huge failure uh I mean we all assume that our intentions matter and that our beliefs matter and that our desires matter and that they make a difference for how we behave but so far all our best scientific accounts don't allow this they pop out at the last moment and say actually all the real causal work is being done by other things right and so so there's always this like danger in the background saying that actually all of our mental life is epiphenomenal all of Consciousness is every phenomenal that that's always there like somewhere lingering in the background and this is where this debate has been stalling for decades so I really just want to like say let's bite the bullet and say you know we need to try something different and maybe it won't work um but you know if you don't even try if you don't even see that there's a problem here uh you know that that's that's uh you know I think at the moment it's unacceptable right we really need to try to do something different here all right so the the two short thoughts um if we don't intend for intention to be a difference that makes a difference all our intentions are on the table and they're all for naught and then the um the second thought is uh the causal efficacy of the map is not causal efficacy in the territory so just saying that a causal parameter in an equation has an influence on another parameter or an equation isn't what one set out to do unless they were doing like forensic equation analysis right right yeah so so I think people get lost in the sense that you can do a lot of scientific work in the old Paradigm right you can happily study Consciousness all you want and say hey I'm Gonna Leave This problem to the other people this is the neuroscientist and the neuroscientists are happily studying how Topline constraints affect all the neuralizing and you know they leave the problem of conscious and intentions to the psychologist and everybody happily publishes a lot of papers and hopes that somewhere along the line these things will meet together um but yeah unfortunately uh you know just saying that there is a kind of mapping is not the answer to this problem of efficacy so just bring it kind of like very clearly let's say um an example that everybody familiar is like a grandmother neuron or Jennifer Aniston neuron or a grid cell neuron or whatever right so we can make a correlational analysis and say the firing of this neuron correlates with you know the person person perceiving this image on the screen let's say or being in this part of the maze or um and that's great uh that's that's very exciting right that we can do that kind of work but then the question becomes well how is that content representational content or Association making a difference for how that neuron is firing and then well what are we going to say there I mean like we can look at that neuron all we want we can study it and it's you know down to the nanoscale if you want and

there's never going to be any place where you can say oh now I understand how Jennifer Aniston makes a difference to this neuron and like it's just not going to happen you're going to have you know chemistry you're going to have electrical potentials you're going to have fluxes you're gonna have membranes you're gonna you know all this messy stuff that you know the the neurobiologists look at but you're not going to have a point where you say oh I see the desire here or I see the value here um and so to just say that these neurons represent a location in a maze has absolutely no efficacy at the level of describing how that neuron behaves and again as long as we don't even see this as a problem I think we are lost because we know we can publish all the papers we want and say hey this part of the brain correlates with this and so on um but the fact that it has content mental content associated with it doesn't appear ultimately in the scientific account of how the systems behave at some point it disappears at a higher level it's there we can say looking from outside the system we can look at what's happening in the environment we can ask the subject and we can look inside the brain and we see all these links very nicely all right Learners forget about these links let's go deeper into the brain and like look at all these you know Cellular Systems and so on oh but what happened to all this other stuff right at some point you're just magically vanished and you know it's a kind of Disappearing trick because we don't even ask the question how did all this extra knowledge that we were positing at the higher level make a difference to what the neuron is doing here all we've got is electrical potentials let's say um yeah and so that's the current state of the art where we don't even see that there's a problem here that there's a gap here um and so uh some colleagues hooked on mine call this the hard problem of content like you know how is it even possible to say that there is content associated with this kind of neuron uh you know which could be true or false could be correct or wrong could be you know different levels of accuracy and so on so all these normative conditions are completely missing from the picture of how the neuron works and so what I'm trying to say here there's an additional problem which is that even if we could give an account of how it is that these neurons also have mental properties like pairing mental content there's an additional problem to say how could that content make a difference to how those neurons behave right now we're talking about the neural level but it's the same kind of issue again like how do we move from something that's normative to something that makes a difference uh you know empirically speaking in the material basis of it so yeah um again deep deep issues uh and the eruption theory is just like a small stab in that direction but I hope that uh what it can do at least is get people thinking uh that you know there's a really interesting problem here that we've been blind to and we should

pay attention to so let's get into eruptions Theory then um and it has a couple of of axioms uh um I think the idea of this so these axioms are supposed to be uh General and intuitive enough that hopefully everybody can agree with them okay so if you don't agree with them let me know because these are not supposed to be controversial okay everybody should be just saying yep okay I can sign off on this so Axiom number one the motivational efficacy and agents motivations as such make a difference to the material basis of the agent's Behavior okay so that's like I want to make a point by lifting my finger okay my finger just went up great you know that it's as trivial as that I'm just saying that you know whatever intentions we have we can see that it has material consequences in the world this is his everyday experience our whole life world is based on this that's him right we didn't believe in this you know it was like the whole all of social order would collapse somehow I mean we just assume that there's a relationship between people's motivations and intentions and how they behave all right so that's Axiom one Axiom 2 brings out the problem here which is that it has intention with our best science so far so action 2 says incomplete materiality it is impossible to measure the scientific instruments how motivations as such make a difference to the material basis of behavior and so this is what we've been talking about you know no matter how close I study how the brain or the body works and you know put it under the microscope put it under uh whatever the scalpel open up your head look inside you know your brain whatever we never see motivations as such making a difference to what anything behaving it's just material processes so for most people this is where the real problems this is the problem of free will this is the Mind Body problem this is the problem of mental causation you know this is the problem of you know can there be the cause of closure of Nature and like how does the Mind fit into the natural order all of that is impact you know in these two axioms which each of them I think we should accept so I'm not saying we should reject them what I'm saying is that we should live with this tension and not try to immediately jump to a conclusion that would remove it both of these axioms are independently valid now the question is how do we deal with the tension between them and so a first step in that direction is to introduce another Axiom right so action number three under determined materiality an agent's behavior is underdetermined by its material basis and this is crucial this comes out of the the discussion we had in the beginning that from a dynamical systems point of view everything is predetermined by the laws of the Dynamics and the initial conditions no surprises right so I could always rewind the tape and know exactly what's going to happen uh and that's to some extent the kind of the concept of nature that a lot of inactive but also cognitive science in

general uh works with it's like we're asking these hard questions about how to fit Axiom one and two together in the context of trying to fit it into a Clockwork universe that you know was in fashion at the time of let's say Newton and LaPlace but it's long gone and the rest of the Natural Sciences right so why are we forcing ourselves to try to solve these ultimate problems in the context of a physics that's you know century is outdated you know that that doesn't make much sense to me so I think this Axiom here might be questionable for some people who really think that um you know science should be in the business of recovering a deterministic worldview and so in for them even quantum physics is not an argument against it because they believe that some future development of quantum physics will show that everything is deterministic after all like a kind of Einstein's you know goes down and play dice kind of view um but I think that you know why why should we want that uh you know it it feels much more natural to say that nature is messy uh uh and like you know there's degrees of freedom that can't be always closed and so on so again this this actually might be the most controversial of the three but I think in terms of our best science it's actually relatively consistent with what we know about nature there are aspects of nature in which we cannot with absolute Precision predetermine what will happen next even if we knew everything about the universe at that moment and in this past all right that's for all of nature that doesn't say anything specific about us and how we interact with the world but that's just to say that the only reason why Axiom 2 looks like its intention with axiom1 is if you assume that Axiom 2 also gives you a complete account of nature right if it gave you a complete account of nature there will be no room for action one if Axiom 2 that we don't find Direct measurable evidence of motivation making difference all right but at the same time we know that action 2 also doesn't give us the full picture of what's going on all right then the tension is slightly diffused because there's space for something to make a difference which is not accounted for in Axiom 2. all right so it's kind of just like opening the window of possibilities just logically speaking for the moment all right so those are the the three axioms now let's see what does eruption Theory make of this um so they kind of breakthrough conceptual breakthrough that that occurred to me in terms of getting serious about how something like an Asian level property could make a difference in the material basis that is described by Axiom II while not violating accept two which means that my motivation involving account can't just say hey there are motivations inside your brain okay that would be this kind of going back to the diagram maybe that we had in the beginning where there's a value system inside the system somewhere making judgments right so that's kind of like placing the value inside the organism but it can also not just be an emerging

property like an order parameter which was kind of like the the right hand side of that diagram where value is this kind of emergent uh property neither of those works for me neither of them gives me a motivation involving account basically they substitute motivations for something else whether it's a local or Global property of the system so what would a motivation making a difference in the material basis actually look like can't appear as a motivation because it's not I mean it's it's not it's not possible to ask the question of nature by measuring it and then say no the answer is not three it's a motivation it just doesn't work like that right and so what how can it appear well it must appear somehow as part of the material basis but it's going to be a very strange kind of object because it's not really an object because it's normative because it's qualitative and so what I want to do is flip this on this head like what uh the motivation does is underdetermine the material constraints it basically means that let's imagine that I'm recording from your brain and I got like you know my time series a signal um you know recording and all the kind of material organizational properties of your body and the environment are all constraining how you know your electrical potential is changing over time now if I wanted to say that something is happening in your signal here which is because of a motivation making a difference to how you behave well that motivation cannot be captured by the material organizational constraints directly it's something else making a difference and something else making a difference from the point of view of the material organizational properties of the system will look like noise it will look like something breaks in the system that's unpredictable something that is unintelligible like no matter how much I study the system no matter how much I describe it understand it something just happened in my measurement that I cannot fully account for no matter what Precision I have no matter how much I know about the history of my recordings and that's because it's the motivation itself that's making a difference and it cannot be captured at that level of description um the way I put it here as a working hypothesis is that the more an agents and body activity is motivated the less that activity is determined by its material basis right because other things are shaping the activity and they are not identifiable with the material basis because they're motivations because their desires because their beliefs because they are wants um and so although all of those have a material basis they are not identical with that material basis they make a difference but that difference cannot be reduced to a difference just at the level of a material basis and this might sound a little bit strange but uh to some extent um it fits quite well with certain Trends in the inactive approach to link the philosophy that it does with Eastern style of thinking so in Japan for example we have a a school of Zen which has a nice phrase which was very popular for virula

for example that says if you think about the relationship between the mind and the body um well it's not one okay so you can't just collapse them but it's also not two you know so a complete Independence the right way of thinking about the relationship between the mind and the body is that it's not one and not two right and that's it that's all they have to say it's not one and it's not two and so it's it's a kind of strange intermediate thing um and that means that there are trust order effects their interaction effects one can make a difference to the other but without collapsing into the other um and so to recognize when this happens when does the one side reach into the other and make a difference without becoming the other side that difference will look unintelligible from within that side is something they cannot grasp on its own terms because it's coming from somewhere else with a different kind of specific qualities in terms of its ontology and I mean this is all very abstract but like you know uh when we go concretely now um it leads itself quite nicely to quantification because we do have measures of surprise we do have measures of unpredictability we do have measures of noise in the system um the the way in which I propose to as an approximation to to measure eruptions is just through entropy right what they do what is happening is that your motivation is causing a certain kind of disorder in the system disorder in the sense that purely material organizational constraints are bracketed for a moment which is the same as saying that they're having less of an ordering influence on so it's a disorder if you can measure the entropy now in the classical libertarian work on this you know once you get rid of determinism there's always this danger of like well doesn't that mean that everything is now random like how would that be useful right um but you know from the point of view of complex adaptive systems that's not a worry anymore because actually we've got lots of interesting work that shows um breaks you know going back to Ashby actually the source of adaptivity you know that they're what allows the system to be flexible uh and to to adapt to its environment that's what is exactly what is needed in order to be flexible and so far from saying that what you said in the beginning like uh earlier that variability is actually a sign of Health right so at to some extent you know it's not bad that the way in which our motivations make a difference to our body is by destabilization by disordering by bracketing constraints that's exactly the kind of degrees of freedom that our bodies need in order to be flexible and healthy so it's a very nice fit with what we already know actually we know that these are markers of consciousness of agency of Health but now what we have is a kind of larger framework that explains why is it the case that let's say neural entropy tracks levels of awareness oh that's great right no not not it fits very nicely now we can it's not just no longer empirical work that says well you know we're measuring this and

we're correlating how much entropy with whether you're asleep or comatose or in a psychedelic state but actually we can say well you know giving our a priori principles the axioms that I outline we expect that when these things affect each other given that the axioms cannot be violated this will Express itself as a certain kind of perturbation or disorder appearing in our measurements and that's what we find and so now we have a really nice theoretical framework that can actually explain to us why our variability and entropy associated with agency and with health and so yes we'll try to do a little restatement and and see if I'm on the right track here so I'm imagining a person in a device that can move their arm so in one situation where this device entirely moves their arm for them 100 of the variance in movement is explained by external observables and so whether now they may have not had a motivation to move it or maybe they really did but if that would be um Epi phenomenal because it basically didn't make a difference the work was all done by the external apparatus um so we can say 100 of the variance was explained what is every single situation other than that well it's less than a hundred percent of the variance explained so there's some difference there's some you know r^2 that's between zero and one and in all like real living and breathing situations it's a lot less than one so there's some space where um the uh material basis even if we believe that we're fully measured wouldn't fully explain it exactly the second piece that I see happening is motivation at least in in our lives at least in my to-do list it feels like an ordering principle and and there's like a way or there's a sense that it is an ordering principle but especially in connection with Ashby and the brakes motivation could be an Adaptive disordering principle that actually supports the movement to Adaptive regime order exactly yeah so it locally disorganizes but then at some meso scale or perhaps some some macro scale it provides an ordering and um as per that first apparatus thought experiment the extent to which motivation is disordering um the space it has to play is in the Unexplained variance by the material consequences that's exactly right yeah yep you got it so let me just unpack that a little bit more so good that you you prompt me um you're right so we are not anymore on this view directly in control of our behavior and that's a little bit disconcerting at first to to Grapple like what we can do to our bodies is set the conditions we can open spaces or uh we can open degrees of freedom we can open spaces of potentiality but we have to trust our bodies and their environments to fill those spaces in the right way all right so it's like I'm opening a space and say voila right now is the possibility for a new Behavior to emerge or to behavior to switch to something else but what behavior is going to be selected that's not directly under our control that will depend on our history our predisposition our body

capacities our body memory it will depend on the affordances of the environment what kind of interaction process are we engaged in and so suddenly all of this work and embody cognition fits in very neatly all of that suddenly comes into play and says actually action self-organizes uh based on all of these criteria of historicity and then body man and situatedness what we do as agents is open spaces for those processes to take place and that's our agency and then of course we kind of upload the rest to our bodies and as long as everything goes well our bodies are actually quite capable of responding in the right way but that's something that's acquired that's not something that's pre-given so to speak we need to undergo the right history to acquire these dispositions and these Tendencies such that when we're in a situation and we open the space the space is filled in the right way and that's very interesting this kind of like it's got a mixed concept of agency that we're partially in control but like just the last bit excludes our graphs we can't force it um but I feel actually that to some extent that's kind of what experience feels like like I can I can try to make this point by speaking and then I kind of find the words most of the time but I just have to trust my body that the right words come out I cannot choose the words in advance and the same is true of all of our actions like you know I can make an intention to reach something but sometimes I misreach you know and for some people who have a body that's not quite optimally organize reaching can be a big problem actually right so so we really are kind of dependent on the body to fill these spaces in the right way and affordances are a big part of this too so um you know there's a general idea that we should choose our environments carefully right because they will shape the way we behave and think and so on and yeah it fits again perfectly with this story that um we open the spaces but then the environment will solicit certain kinds of propensities from us and we're not always fully in control of exactly what it is that will come out of us um and um so it's a kind of compromise between a kind of libertarian point of view a full control over our behavior and a kind of deterministic point of view of completely lack of control of what's Happening where yeah we can decide to open a space of possibilities but actually it's other forces that close the space awesome please continue many many interesting angles yeah I mean uh that part I skipped it but good that you brought up um is described also in the in the paper in terms of a thesis of Attunement so it kind of goes with this that uh um I have three thesis here one is the eruption thesis which says that if they are cross-border effects we will see them as a disordering consequence in the domain that's being affected um because it's coming from outside of that domain and then if that's the case we have this challenge in front of us how do we explain that this this order actually does some good work for the organism uh and one

thing there is that there are two possibilities of this order one is that it could be a whole system consequence so you know there's a lot of work looking at Silver inferentiality uh and Barilla himself tried to link also quizzes with girdle's work on the incompleteness theorem and so on and so maybe actually eruptions are system level consequences of self-reference and incompleteness at that level but you know that's further work or it's something that happens very locally that you know at the smallest micro scale there's a kind of glitch in the system and then permits kind of like a disordering of some you know particle configurations or something like that basically um you know using up some energy there uh that's also possible if that's the case we would have to have a scaling up effect so how do we go from a small disorder that is barely even measurable or maybe not even measurable because almost almost infinitely small perturbation to something that makes a difference at the level of the scale of behavior right but again we've got a whole nice body of work in artificial life to slot in there which is that and the organisms are organized so to be sensitive to very small differences right so this is a self-organized criticality we can talk about chaos we can talk about scale free Dynamics so there's a whole body of work that already shows us that the body is organized so as to macro to amplify very small perturbations that's nice now we again we have an interesting idea of like why is it the case that the body is organized in that way in this kind of poised way and then if you do have eruption scaled up to the macroscopic level still there would be the sense of like you know that that's not useful really to have a huge disorder washing through your body unless you're doing it in the context of being attuned already to the environment in the right way in which case actually it will become a source of flexibility such that you're no longer trapped into kind of committed coping like you know uh um you know I have to continue like highly ears Hammer right I have to hammer I have to hammer like you know my house is burning down but I have to hammer I have to hammer like you know it would be very bad if we are really good at doing a single task and could never switch to another task and so how how open are we to switching tasks um that openness the switching task can be regulated by the eruptions of the microscopic level but like I said like exactly what action will happen next will depend on how we are attuned to the situation and so this achievement then brings in this whole working in body cognition of how a lot of our action selection happens in the uh organism environment interaction process it makes me think about Reservoir Computing and about the the environment as a dampener for the eruption because if if we do grant that there is endogenous excitation so it's not just an outside in the car the computer world so we we are going to a grant that there's some kind of endogenous generativity um or spontaneity then

when it interfaces with the environment there's basically two options from a dynamical systems perspective either the environment could dampen the excitation like you could yell and then that could dampen or it would be a positive um feedback and it would be like a spiraling out of control at which point that coupling wouldn't um would basically escalate until it reached some other negative feedback point so the environment plays a a critical dampening role through Attunement so so that the um so that that break can find a new non-equilibrium steady state or A New Path right so there's a kind of question here of how what's the right balance for how much eruptions do you need in order to you know be an optimal situation so as you said like there could be situations when it happens too much right that you don't ever have time to properly settle into a new attractor or into a new activity um and maybe we could think of some disorders like schizophrenia in this way right there's so much you know the same motivational involvement and what's going on all the time hyperreflexivity that you're basically destabilizing you know the system constantly and never allowing it to properly settle and there's interesting links there also with the fact that there you know there's their body memory gets lost you know like uh and so on so um there's that it can go to in too much Direction and also there's the other direction that says what if it doesn't happen enough you know a certain kind of rigidity will set in you know you will not be able to switch context easily you will not be able to deal with novelty easily um you know and maybe something like depression could be something in that direction that you know like you feel like the world's closed off to you there's no longer any possibilities for change or development um and that brings then like you know this this whole work into another Direction which is can we use it to better understand how to treat people um you know so what if that the schizophrenic what can we do for them to kind of like dampen more maybe the eruptions that they're constantly sending into their body versus for the depressed patient maybe it's the opposite how can we make them more susceptible to being open to eruptions and interestingly it can be both ways right so maybe in the case of depression it's something that's happening at the level of the person level the Asian level experience level at the middle of motivations and the body itself might actually be fine to some extent that's an interesting possibility here right and then in that case we have to work on the level of their experience how do we get them more motivated to some extent and then that will kind of kick-start things or on the other hand it could be a physiological thing right you know the at the person level everything is fine except that the brain maybe for example in Alzheimer's and Dementia is crossed it over and it's no longer susceptible to these degrees of freedom being injected into it right and so what looked from

the outside as two cases of becoming maybe more rigid and less responsive and less flexible might have opposite sources in terms of does the disorder lie more at the the call side of the eruptions or the consequence side of their options right and so we can think about you know this now because previously these things were collapsed right we couldn't easily differentiate them but now we have these two possibilities to look at this like um and a lot of the interventions that do work in the clinic they say for depression for for example kind of feel like they're almost like simulating eruptions right so why does psychedelics work so well for for depression well because it's a huge destabilization of the system right and it works both at the motivational level that you have a huge increase in the degrees of your cognitive processes of your experiential processes which then also you know like destabilizes the system at the brain level um or another example uh obsessive compulsive disorder uh and you know why does something like deep brain stimulation help um or electric convulsive therapy in other cases like you know for schizophrenia why is it that these seemingly kind of crude interventions of just injecting a lot of arbitrary noise into the brain which is irrelevant from the point of view of what the brain itself is doing right so it's undergoing its processes and suddenly from the outside we inject you know like these hard uh you know activity that doesn't is not contextualized it doesn't fit in with what the brain was already doing and yet what happens is that it actually makes people feel better it makes them feel refreshed it makes them feel more stable it makes them feel more flexible why why should the injection of these kind of like breaks of activity have these consequences and so with a Russian Theory we can actually start to conceptualize this in a different way and think that you know what we're doing there is maybe amplifying or substituting you know uh eruptions artificially uh and and the body is there being responsive to them and then once things are like more you know let's say well oiled again the person themselves can like you know take over the role again of of you know doing eruptions themselves rather than having it induced from from the outside and it makes me think of the kind of unreasonable efficacy of just general physiological interventions like sleep walking eating cold showers meditation like why does it all why do all roads kind of Channel back to these large-scale breaks because they um at a a deeper nested level allow the adaptivity to exercise especially the exercise one you know there's a recent paper and and been asked about you know measuring neural entropy in a huge fmri data set and then ranking like what's the highest level and actually motion-based tasks have the highest level of neural entropy associated with it and you know for people doing EEG you know we kind of know it because you know it's a nuisance right you say how can I do my EG and I

have to like keep people sitting and not moving because if they start moving my signals go over the place and you know that a lot of that is muscle artifacts but you know in general the idea is clear right like the more you you move the more things are happening uh in in your brain and so um yeah just going for a walk uh you know actually uh could could really make some differences yes many I mean this is the heart of um of the inactive Quest and of active inference which is if we only wanted to describe the special case of active inference where um there was no action or where actions were being caused by something that didn't matter then it would collapse back onto passive inference but the challenge of scientific analysis of behavior is actually uh serious and I think that you're pointing to some uh kind of figure and ground type conceptualizations such that what has been thrown into the variance unexplained category um the motivation was getting thrown out with the bathwater that's right yeah that's a very nice way of putting it and um maybe just a final thought here and that has to do with the replication crisis you know something just to ponder right like why is it that in Psychology it's so hard for people to behave the same way and experiments you know and and we try to counteract that by having larger sample sizes and more control conditions and so on and you know if you can't replicate you we we think that you know I guess it wasn't controlled properly or we didn't have enough sample size and so on but what if all that variability is just actually a positive signal that someone you know took the task seriously and made a difference to what's going on and and people are not like machines like you know we can't control directly how we will respond to the situations what we can do is open degrees of freedom and then let the situation you know help us self-organize the action and so of course if you fully control the situation such that you know there's only two action choices or something like that and you know everything else is also pretty like fixed well then you can get people to behave like you know like they are reactive systems almost and then things are replicable but you might just lose completely what's interesting about people in the first place and so we might have to think turn things around and just say how do we operationalize this variability in such a way that rather than throwing it out or trying to control it away we kind of like take it as that's the marker of the person actually making a difference here in the task and you know like in physics similar Revolution happened right there was a lot of annoyance before quantum physics came around of like why are we getting this interference and so on and it took a while for people to accept that actually the way forward is to just accept this uncertainty and operationalize it and work with it and get equations to work with it and then suddenly you know everything you know we've got the most precise signs in our hands and but it had this

you know this is this strange switch where we suddenly had to realize that the kind of uncertainty was not an artifact of us uh you know not observing things in the right way but it was a property of the phenomenon itself that we were trying to understand and I think we need to do the same thing for people and agents more and this is the classical anomaly in the Kuhnian Paradigm change which is it's not that normal science is unaware of the future Paradigm it's that it has a error variance explanation we made the partition between signal and noise as we know it today and that's noise which means we can describe it with the gaussian which means it has all these nice mathematical properties and so so it has this kind of empirical feel of a total statistical model which can be convincing but as were the total statistical models of epicycles and pre-quantum photoelectric phenomena yeah you're right so and I think one of the things that uh I mean people I think were always frustrated by you know the large error bars and so on right um and we can we can handle with them but you know we prefer not to have them um but I think the switch wasn't possible yet because we didn't have a good grasp of why there should be this uncertainty like what explains it where does it come from when should we expect it and when not under what conditions and how can we formalize it and operationalize it and measure it so I hope eruption Theory points a little bit in that direction by saying hey for example if you look at neural entropy in the brain maybe it's actually you know partially picking up on the subjective involvement of making it that's making a difference here and how the behavior or the brain you know processes activity in that case you know um it's not just that we're anymore like looking at the error bars but now it's becoming something in itself that's of interest to us and that we want to study and understand so I hopefully you know it kind of like can and you know facilitate the power because if you don't have another alternative well you're stuck with just the error bars right but yeah I think we're seeing like the light at the other end of the tunnel that maybe we can head towards from now on um one joke and then I'll read a comment from the chat and then you can give some closing thoughts uh the joke uh formerly known as error bars now known as motivation bars okay I'll read it I'll read a comment from um the chat and then you can give any just any reflection or any closing thoughts or next steps so um IA Roland Rodriguez wrote I think uh if all behaviors described as a gentle are just products of a causal chain of interactions and exchanges in and out costs then there isn't this free leap from nowhere the modeling of organisms doesn't seem to be able to account for this Gap in the mechanical unfolding of energy flow where an a priori is not an empirical sedimentation and habituation Pere also means to impose so one in one question is is life hard to model because it imposes priors seemingly from nowhere as regards causal

Dynamics within the infinite encapsulation of blankets I.E like the opening question regarding the reduction of action to Mechanical motion is the imposition of vital priors hard to account for in mechanistic immersionist modeling wow there's a lot to impact there now let me think so there was a mention of uh energy so this is something that uh I'm very keenly working on at the moment and I think I just mentioned in passing but if we think about eruptions as a disordering consequence then you know let's say uh this is just a kind of thought experiment but let's say that you know we can look at a particle moving through space uh somewhere in your body as part of some sort of material flow right and that eruption happens at that point and that location um what would it look like from the point of view of measuring this particle well suddenly it would become unpredictable right instead of just moving forward it was just move in some arbitrary configuration because it's no longer constrained or only constrained by the kind of forces that kept it straight in the beginning and that's basically an increase in disorder from the point of view of the system which means that whatever energy was contained in that particle now is no longer available for for doing work um and so one interesting implication which is still totally crazy and I'm even almost scared to say it is that mental work costs energy just like you know physical work or mechanical work and both of them cause work as cause energy but not for the same reason so it's not that mental work costs energy because ultimately it's just another form of mechanical work it is its own drain on energy according to this idea that might be a very small change you know maybe it's not even detectable or it could be a big one who knows I mean you know the the brain is the most energy expensive tissue of the entire or like body so who knows like whether all the energy budgets add up the way that we expect them to I guess that's like you know some interesting empirical uh hypothesis that could be you know somewhere looked at and what was the other one about the about the imposition and the priors and so on so yeah so I've kind of left out the origin story a little bit in the European series I mean I gave the kind of an active story of why should we think that organisms have an intrinsic perspective and why they're different why living beings are different from other kinds of power from equilibrium um but I actually want eruption Theory to be independent of that a little bit you know you know that's my preferred way of getting to an agent you know for me that's in a compelling story but you can have any kind of story you want uh and you could still plug in eruption Theory at the point of saying now that I've got an agent how does that agent make a difference to the material basis that's giving rise to it and then eruption C we can help you understand those those links so um I guess one more way of thinking about this in terms of a more cosmological side is that if

eruptions are on the right track and are actually a genuine source of entropy in the organism that can't be reduced to other sources of entropy well it increases entropy production so that means life due to its active regulation due to its agency produces more entropy than non-life in the sense of other far from equilibrium systems like a convection role or something like that which is a popular example for let's say ecological psychology to show that increased heat transfer in actually has organizing Tendencies you know so if you have things rolling like this heat can flow more easily from the pan into the air and top okay but if it's living and it's actively regulating those flows it will produce even more entropy and so if we think of the general tendency of the universe towards an increased entropy production then living systems become actually kind of natural consequence of that unfolding in the sense that they're adding another level of complexity on top of other far from equilibrium systems that is another source of Entry production and so yeah it's kind of like that would mean that life is kind of expected or you know spontaneous rather than something that is rare and weaker than the universe while the closing image for me is I'm watching the water boil and it's already far it's rolling but jiggling it and and just helping it boil and having that just Channel energy right so that's a yeah that's a metaphor for what life is all about like you know what we're doing is just like you know channeling the flows well thank you very much for this excellent guest stream I hope that the work is well received in your Journeys and please always feel welcome to um come back when the time is right okay great thank you so much for having me that was a lot of fun awesome thank you Tom farewell all right thank you bye-bye bye